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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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1. Team & Project Overview

Team Name	奶龙
Project / Solution Name	Nexus
Name (Leader)	LING YUXUAN
Email & Phone (Leader)	1730211684@qq.com 1123232813
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://nexus-nailong-2026.d7ystszcjf.chatgpt.site/
Code Repository URL (public / access granted to judges)	https://github.com/junzhanglee/SDC2026-domain2-nailong/tree/feat/nexus-mvp

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Sites hosting serves the browser interface. Local business rules lead to human approval, task creation and activity logs stored in localStorage. Selected messages go to a Sites-hosted Cloudflare Worker, which checks ChatGPT sign-in and D1 daily usage limits before calling the OpenAI Responses API (gpt-4.1-mini). Validated drafts return for human review; successful generation is currently blocked by insufficient API credits. D1 stores AI usage counters, not business records. Local Excel/CSV files are parsed in the browser using fflate. SDC-provided Replit premium access supported development/testing; the live prototype is hosted on Sites.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
HTML5	Web page structure and accessible forms	N/A — web standard.
CSS3 / modern CSS	Responsive layouts, styling and glass effects.	**N/A — web standard.**
JavaScript ES Modules (ES2015+)	Interface interactions, business rules, approval workflows and browser storage.	**N/A — language standard.**
Node. js v24. 19. 0	Development tooling, automated tests and static-site builds.	MIT License

Name & Version	Purpose in Project	License
fflate 0.8.3	Browser ZIP decompression for local Excel (.xlsx) imports.	MIT License

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
•OpenAI Responses API — GPT-4.1 mini	Extract customer requirements and generate editable reply drafts for human review. Integration is deployed; successful generation is currently blocked by insufficient API credits.	Paid (pay-as-you-go); not sponsor-provided

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
OpenAI Sites	Website hosting, deployment and server secret management.	Managed hosting / Region not verified
Cloudflare Workers (via Sites)	Runs backend endpoints and securely calls the OpenAI API.	Platform-managed / Region not verified
Cloudflare D1 (via Sites)	Stores daily AI usage counters. Business records remain in browser localStorage.	Platform-managed / Location not verified

Service & Provider	Purpose in Project	Plan / Region
ChatGPT sign-in (via Sites)	Authenticates visitors before allowing AI API calls.	Platform-managed / Region not verified
Replit	Development/testing with SDC premium access.	Sponsor-provided; region not verified

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
GPT-4.1 mini	OpenAI Responses API	Pre-trained; customer requirement extraction and editable reply drafts. No fine-tuning or self-training.	Proprietary model; subject to OpenAI API service terms.

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Nexus demo business data: staff, customers, inventory, leave and tasks	Team-created; repository `apps/web/assets/data.js` and `customer-service.js`	Synthetic	Apache-2.0	N- fictional records

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Workflow and import test fixtures	Team-created; repository `tests/`	Synthetic	Apache-2.0	N
User-entered messages and uploaded Excel/CSV data	Provided by users during application use	Private / user-provided	User-owned; permission required	Y — may contain names or messages
Authentication identifiers and AI usage counters	Sites authentication and Cloudflare D1 `ai_usage`	Private / application-generated	Subject to provider terms and privacy requirements	Y — user identifiers

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
OpenAI ChatGPT — hosted service; model/version not recorded	Product planning, requirement analysis, interface design guidance and submission text drafting.	Major
OpenAI Codex — desktop app; version not recorded	Frontend and backend code generation, debugging, automated tests, API integration, deployment and documentation.	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team refined AI-generated code and text against project requirements. We used 61 automated tests and browser checks to verify business rules, permissions, approvals and Excel imports. We disclosed incomplete integrations and the API credit limitation. Final review, submission accuracy and responsibility for the solution remain with our team.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	SDC-provided premium account access	Used to import and run the Nexus project in a cloud development environment.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit's sponsor-provided premium access supported cloud-based development and testing. It was helpful but replaceable: the project can also run locally and is currently deployed through OpenAI Sites. The core workflow does not depend on Replit-specific services.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Nexus stores entered names, profiles, messages, requests and activity records in browser localStorage until users clear site data. Excel/CSV files are processed locally and are not uploaded. AI requests send only the selected message and Malaysia date to OpenAI after an explicit user action. API keys remain in server secrets. AI access requires ChatGPT sign-in; platform user identifiers and daily usage counters are stored in Cloudflare D1. These counters currently have no automatic deletion schedule. Requests use HTTPS and `store:false`, which does not guarantee zero provider retention. Frontend role filtering is for demonstration, not production-grade authorization. Users are instructed to use synthetic data and avoid sensitive information.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

AI may misinterpret messages, invent details, produce biased wording or follow malicious instructions. Mitigations include structured outputs, source-evidence validation, missing-field handling and mandatory human review. AI cannot automatically send replies, approve requests, make payments or create tasks. Authentication, input limits and daily quotas restrict API access. Provider failures retain the original message and show an error. OpenAI is currently the sole model provider, and insufficient credits prevent successful generation verification. Business records remain browser-local, with no cross-device synchronization or production tenant isolation. WhatsApp and Microsoft 365 cloud synchronization are not connected.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ✅ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ✅ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ✅ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ✅ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ✅ Any use of official SDC sponsor resources has been declared in Section 5.
- ✅ The prototype described here is accessible to judges through the links provided in Section 1.
- ✅ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] LING YUXUAN LI JUNZHANG XIE WEIYE HU ZHAOMING

[Date] 24/10/2026